

WhatNext Vision Motors: Shaping the Future of Mobility with Innovation and Excellence

ABSTRACT

WhatsNext Vision Motors is initiating a significant Salesforce CRM deployment designed to improve customer interactions and operational effectiveness in the automotive industry. At the heart of this solution is a data architecture that combines vehicle stock, dealer locations, and customer information. The CRM utilizes automated workflows and Apex triggers to apply business rules—preventing orders for vehicles that aren't in stock and automatically directing new orders to the closest dealer by using customer location data.

In addition to these immediate safeguards, the initiative incorporates scheduled tasks: batch Apex refreshes stock levels for large orders, while scheduled Apex sends out email notifications for upcoming test drives and inventory restocks. Together, these elements create a unified system that provides clear and precise tracking of orders—labeling them as “Pending” or “Confirmed” based on current stock status.

The technical rollout encompasses well-organized data modeling for vehicle and dealer information, the Lightning App Builder to facilitate user-friendly interactions, record-triggered flows for executing real-time business logic, and strong Apex programming for validation and automation. By lessening the need for manual processes, reducing customer inconvenience, and enhancing the speed of stock updates, this project aims to greatly boost customer satisfaction and internal efficiency—driving WhatsNext Vision Motors towards more flexible and customer-focused operations.

OBJECTIVE

The goal of this initiative is to improve the way customers place orders by utilizing Salesforce automation and smart data management. It intends to optimize dealer assignments, verify stock availability in real-time, and deliver precise updates on order statuses. In this way, WhatsNext Vision Motors strives to increase customer happiness and boost operational productivity throughout its transportation services.

- Improved Customer Interaction: Making the ordering journey easier with automated dealer recommendations and instant notifications.
- Instant Inventory Oversight: Avoiding situations of stock shortages through precise inventory checks.
- Automated Dealer Allocation: Distributing orders to the closest dealer based on where the customer is located.
- Streamlined Order Handling: Employing scheduled operations to refresh order statuses and enhance fulfillment processes.
- Intelligent Salesforce Automation: Utilizing Apex, Flows, and Batch Jobs to streamline

and expand operations.

TECHNOLOGY DESCRIPTION

Salesforce:-

Salesforce is a web-based system for managing customer relationships that helps companies handle their interactions, information, and activities effectively. In this project, it serves the purpose of storing and organizing information about vehicles, dealership sites, and customer requests. The system facilitates automation via Flows, Apex triggers, and batch jobs to guarantee precise inventory checks and intelligent dealer allocations. By utilizing Salesforce, WhatsNext Vision Motors improves its operational productivity and provides a smooth, customer-focused ordering process.

Custom Objects:-

Objects in salesforce are like tables in a database. Custom Objects are created to store specific data.

Example:

- Vehicle__c - Stores vehicle details
- Vehicle_Dealer__c- Stores authorized dealer info
- Vehicle_Customer__c- Stores customer details
- Vehicle_Order__c- Tracks vehicle purchases
- Vehicle_Test_Drive__c- Tracks test drive bookings
- Vehicle_Service_Request__c- Tracks vehicle servicing requests

Tabs:-

Tabs are used to display data in the Salesforce UI

Example: A tab for vehicle__c allows users to easily view, create, and manage vehicle

Custom APP:-

An App in Salesforce is a collection of tabs grouped together for a specific business purpose, allowing users to access related data and functionality in one place.

Fields:-

A **fields** in Salesforce is a specific piece of data stored in an object, much like a column in a spreadsheet. Each field holds a particular type of information such as text, date, number, or picklist values.

Example: Vehicle_Model__c – A **Picklist** field in the Vehicle__c object with options like *Sedan, SUV, EV*, etc.

Flows:-

Flows automate business logic without writing code. They can create or update records, and send notifications automatically.

Example:-

- A flow triggers an email alert whenever a new order is created. Apex:-

Apex is Salesforce's Object-Oriented programming language. It allows developers to write custom logic .

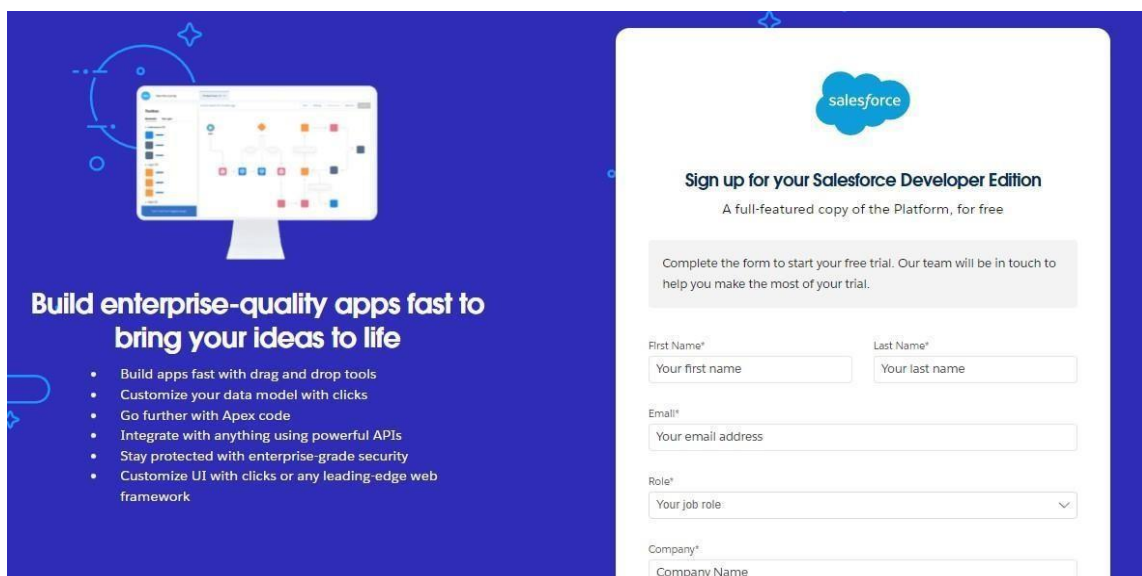
Example Triggers:

- A trigger automatically validates stock availability when a new vehicle order is created.
- It prevents orders if the vehicle is out of stock.
- Once the order is confirmed, the trigger updates the stock quantity by reducing it.

DETAILED EXECUTION OF PROJECT PHASES

1. Developer Org Setup

- A Salesforce Developer Org was Created Using <https://developer.salesforce.com/signup>



- The account was verified, password set , and access was granted to the Salesforce Setup page.

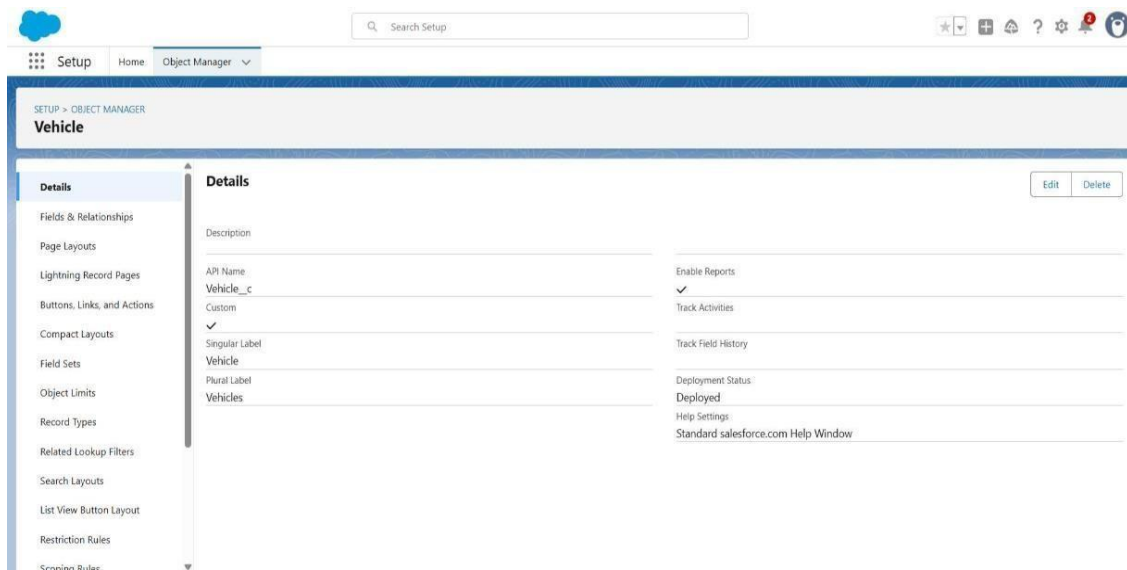
2. Custom Object Creation

Six Custom Objects were created to store business-critical data

- **Vehicle** – Stores vehicle details like model, stock, price, and status.
- **Vehicle Dealer** – Contains information about dealers such as location and contact details.
- **Vehicle Order** – Tracks customer orders, order dates, and order status.
- **Vehicle Customer** – Maintains customer details including contact and preferred vehicle type.
- **Vehicle Test Drive** – Records test drive schedules and status.
- **Vehicle Service Request** – Logs service requests, dates, issues, and progress status.

Steps followed:

- Navigated to Setup → Object Manager → Create → Custom Object.
- Provided Label, Name, and enabled options for Reports and Search.
- Saved and created Tabs for each object.



3. Creating a Custom Tab

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'tab' entered and a list of navigation items: Feature Settings, Analytics, Tableau (with sub-items: Tableau Embedding, Tableau UAF Claims Definition), User Interface, Console Settings, and a highlighted 'Tabs' item. The main content area is titled 'Custom Tabs' and includes a 'New' button and a 'What Is This?' link. Below this is a table of 'Custom Object Tabs'.

Action	Label	Tab Style	Description
Edit Del	Vehicle Customers	People	
Edit Del	Vehicle Dealers	Books	
Edit Del	Vehicle Orders	Alarm clock	
Edit Del	Vehicles	Car	
Edit Del	Vehicle Service Requests	Ticket	
Edit Del	Vehicle Test Drives	Chap	

At the bottom right of the table, there is a message: 'Activate Windows. Go to Settings to activate Windows.'

4. Creating the Lightning App

- A custom Lightning App named WhatNext Vision Motors was created.
- Included tabs: Vehicle, Vehicle Dealer, Vehicle Order, Vehicle Customer, Vehicle Test Drive, Vehicle Service
- Assigned to the System Administrator profile.

The screenshot shows the 'Lightning App Builder' interface. The top navigation bar includes 'Lightning App Builder', 'App Settings', 'Pages', and 'WhatNext Vision Motors'. The left sidebar shows 'App Settings' with sub-items: App Details & Branding (selected), App Options, Utility Items (Desktop Only), Navigation Items, and User Profiles. The main content area is titled 'App Details & Branding' and includes a description: 'Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.'

The 'App Details' section contains the following fields:

- *App Name: WhatNext Vision Motors
- *Developer Name: WhatNext_Vision_Motors
- Description: An app to manage vehicles, dealers, customers, orders, test drives, and service requests.

The 'App Branding' section includes:

- Image: Upload button
- Primary Color Hex Value: #0070D2
- Org Theme Options: ☐ Use the app's image and color instead of the org's custom theme

The 'App Launcher Preview' section shows a preview of the app icon and name: 'WhatNext Vision Motors' with the description 'An app to manage vehicles, dealers, customers, orders,...'

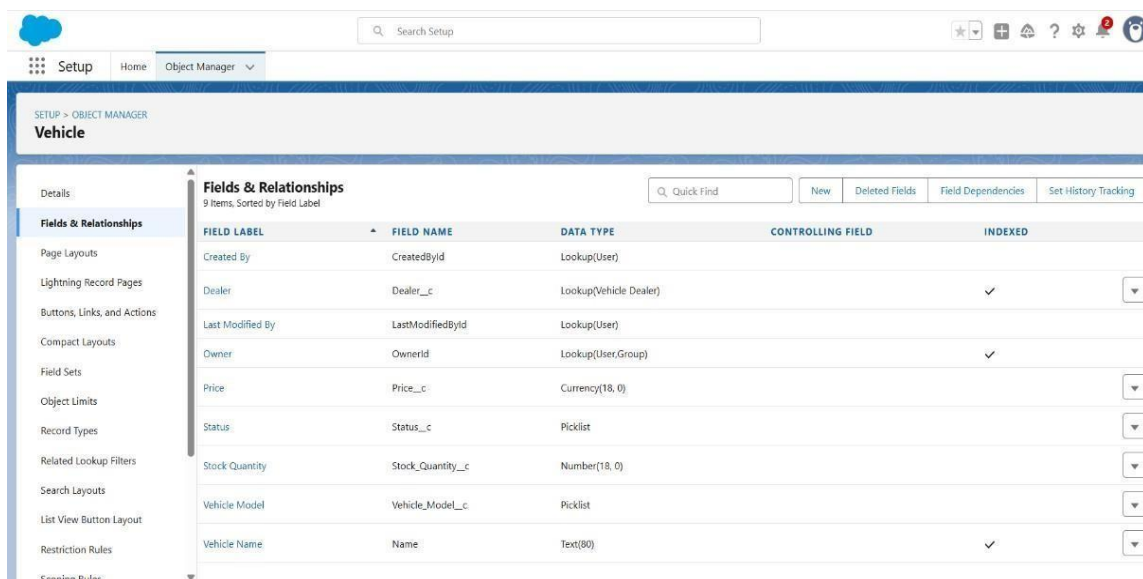
5. Creating Fields

Custom fields were created on each object to store specific business data.

For example, on the `Vehicle_c` object, a Picklist field named `Vehicle_Model_c` was created with values like *Sedan*, *SUV*, and *EV*.

Steps followed:

- Navigated to Setup → Object Manager → [Object Name] → Fields & Relationships → New.



The screenshot shows the Salesforce Setup interface for the 'Vehicle' object. The 'Fields & Relationships' tab is selected, displaying a list of 9 fields. The table below represents the data shown in the screenshot.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Dealer	Dealer__c	Lookup(Vehicle Dealer)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Price	Price__c	Currency(18, 0)		
Status	Status__c	Picklist		
Stock Quantity	Stock_Quantity__c	Number(18, 0)		
Vehicle Model	Vehicle_Model__c	Picklist		
Vehicle Name	Name	Text(80)		✓

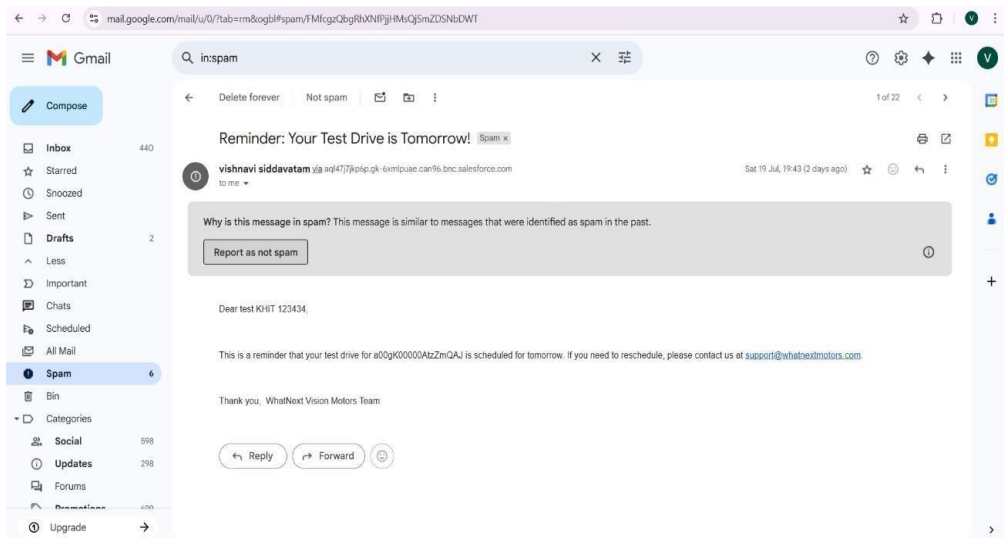
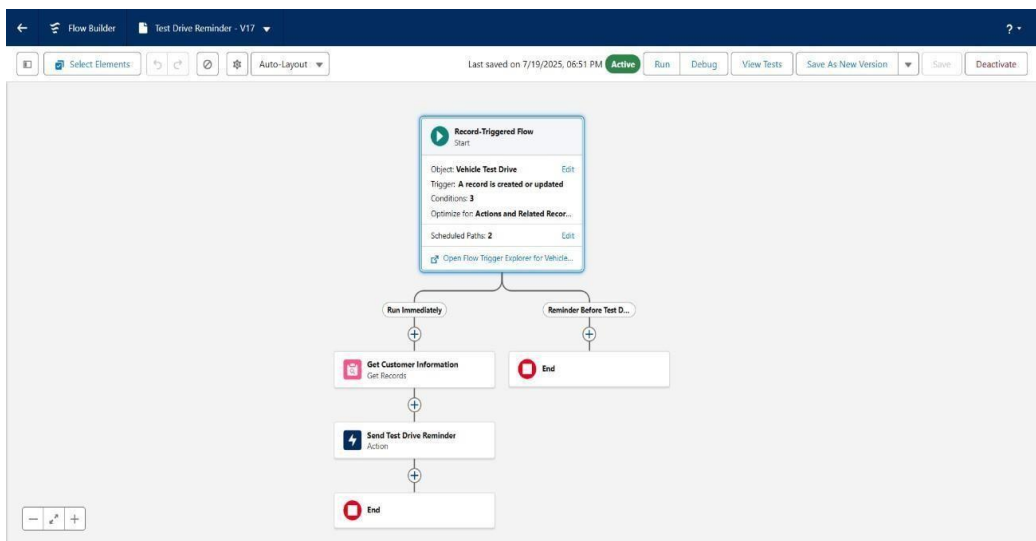
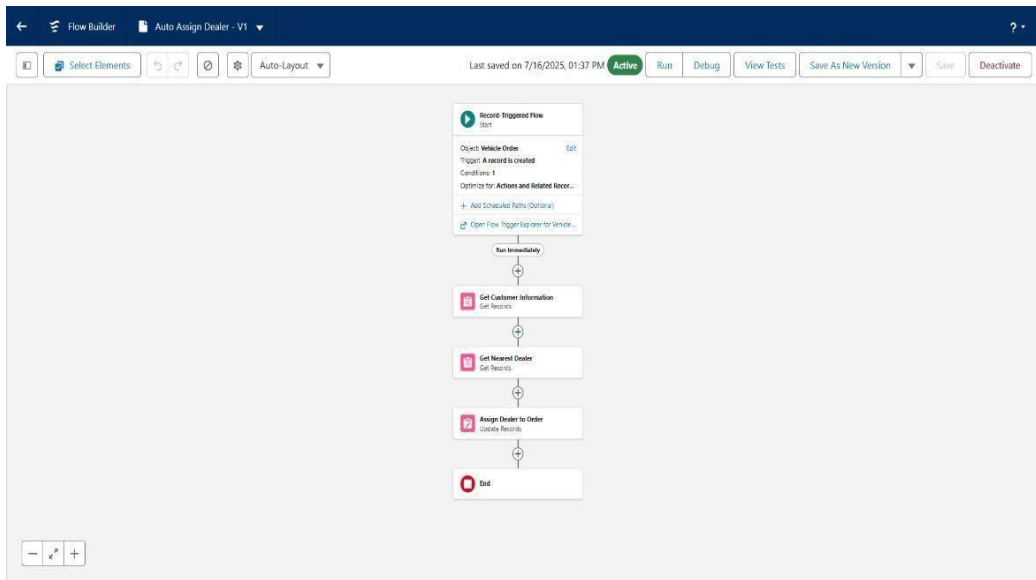
6. Creating Flows

Record-triggered flows were created to automate key business processes without code. For example, a flow was built to **auto-assign the nearest dealer** when a vehicle order is created with status *Pending*.

Another flow sends **email reminders** to customers **one day before** their scheduled test drive.

Steps followed:

- Setup → Flows → New Flow → Record-Triggered Flow.



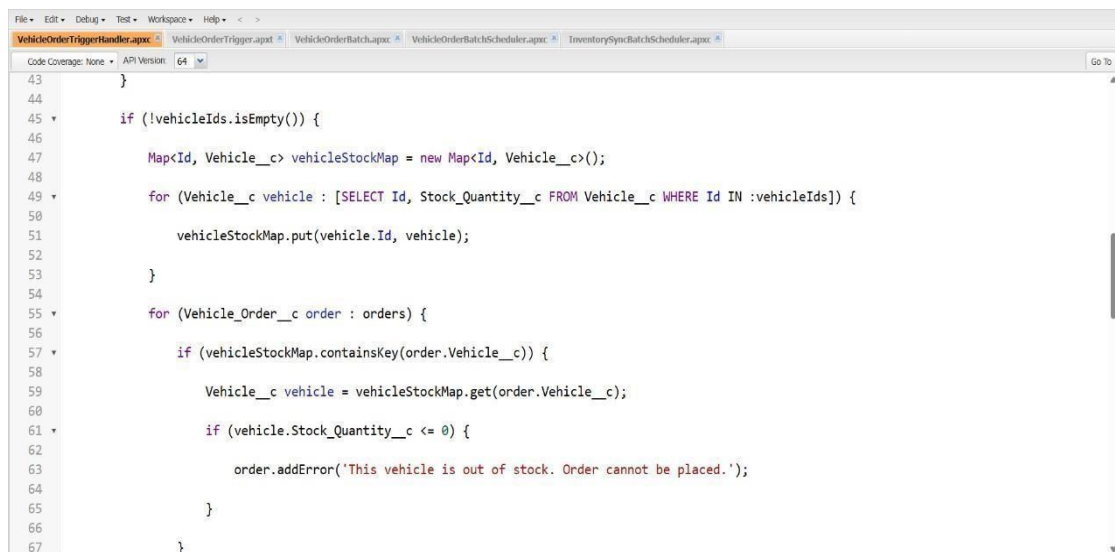
7. Create Apex and Trigger Batch Jobs:

Apex classes and triggers were created to enforce business logic during order processing.

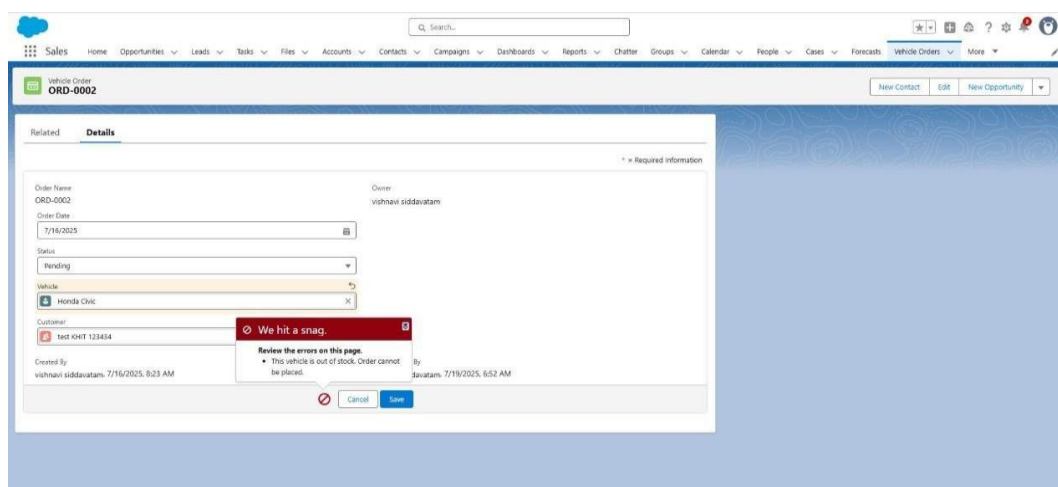
- A **trigger** checks vehicle stock when an order is placed and updates stock on confirmation.
- A **batch class** regularly scans pending orders and updates them to *Confirmed* if stock becomes available.
- A **scheduled class** runs the batch job daily to automate order updates.

Steps followed:

- Developer Console → File → New → Apex Class.



```
43 }
44
45 if (!vehicleIds.isEmpty()) {
46
47     Map<Id, Vehicle__c> vehicleStockMap = new Map<Id, Vehicle__c>();
48
49     for (Vehicle__c vehicle : [SELECT Id, Stock_Quantity__c FROM Vehicle__c WHERE Id IN :vehicleIds]) {
50
51         vehicleStockMap.put(vehicle.Id, vehicle);
52
53     }
54
55     for (Vehicle_Order__c order : orders) {
56
57         if (vehicleStockMap.containsKey(order.Vehicle__c)) {
58
59             Vehicle__c vehicle = vehicleStockMap.get(order.Vehicle__c);
60
61             if (vehicle.Stock_Quantity__c <= 0) {
62
63                 order.addError('This vehicle is out of stock. Order cannot be placed.');
```



Vehicle Order
ORD-0002

Related Details

Order Name: ORD-0002 Owner: vishnavi siddavatam

Order Date: 7/16/2025

Status: Pending

Vehicle: Honda Civic

Customer: test cwt 123456

Created By: vishnavi siddavatam, 7/16/2025, 6:23 AM

By: siddavatam, 7/19/2025, 6:52 AM

We hit a snag.
Review the errors on this page.
• This vehicle is out of stock. Order cannot be placed.

Cancel Save

8. Scheduled Jobs

The **Scheduled Apex Jobs** page in Salesforce displays all Apex jobs that are scheduled to run automatically, such as batch jobs, schedulable classes, and queueable jobs. It provides key details like job name, type, status (Queued, Processing, Completed, Failed), submission time, next scheduled run, and the user who submitted it. From this page, administrators can monitor job progress, check for errors, and manage jobs by aborting, pausing, or deleting them when needed. This helps ensure that automated processes run smoothly and any issues can be quickly identified and resolved.

The screenshot shows the Salesforce Setup page for Scheduled Apex Jobs. The left sidebar contains navigation links for Setup, Home, Object Manager, and a search bar. The main content area is titled 'Scheduled Jobs' and displays a table of all scheduled jobs. A yellow banner at the top of the table indicates that 2% of the scheduled jobs limit is used. The table has columns for Action, Job Name, Submitted By, Submitted, Started, Next Scheduled Run, Type, and Cron Trigger ID. The jobs listed are: Daily Inventory Sync, Daily Vehicle Order Processing, Metalytics Data Loader Job for Org: 00DgK000006xMtp, Program Milestone Computation Cron Job, and Program Status Update Cron Job.

Action	Job Name	Submitted By	Submitted	Started	Next Scheduled Run	Type	Cron Trigger ID
Manage Del Pause Job	Daily Inventory Sync	siddhantam_vishnavi	7/16/2025, 4:48 AM	7/21/2025, 12:02 AM	7/22/2025, 12:00 AM	Scheduled Apex	08egK000007WIP45
Manage Del Pause Job	Daily Vehicle Order Processing	siddhantam_vishnavi	7/16/2025, 4:13 AM	7/20/2025, 12:02 PM	7/21/2025, 12:00 PM	Scheduled Apex	08egK000007W15p
Del	Metalytics Data Loader Job for Org: 00DgK000006xMtp	User Integration	7/7/2025, 11:28 AM	7/20/2025, 6:24 AM	7/21/2025, 6:24 AM	Autonomous Data Loader Job	08egK000006xvBE
	Program Milestone Computation Cron Job	Process, Automated	7/7/2025, 11:28 AM	7/21/2025, 12:00 AM	7/21/2025, 6:59 AM	Program Milestone Computation Cron Job	08egK000006xvBC
	Program Status Update Cron Job	Process, Automated	7/7/2025, 11:28 AM	7/20/2025, 8:01 PM	7/21/2025, 5:00 AM	Program Status Update Cron Job	08egK000006xvBD

PROJECT EXPLANATION WITH REAL-WORLD EXAMPLE

1. Customer

Registration Real-Life

Example:

A customer, **Elijah Mikaelson**, visits the automobile showroom or website.

In Salesforce:

- A new record is created in the **Vehicle_Customer__c** object with his details such as Name, Phone, Email, and Address.
- **Validation Rule:** Ensures the email is valid (e.g., must contain "@gmail.com"). If invalid, an error is shown.

2. Vehicle & Inventory

Management Real-Life Example:

The store admin adds cars like **BMW X5, Toyota Fortuner, etc.**, with stock quantity (e.g., 5 BMWs and 3 Fortuners).

In Salesforce:

- Each vehicle is stored in the **Vehicle__c** object with details like Price, Model, and Stock Quantity.
- **Validation Rule:** Prevents adding a vehicle with negative stock values.

3. Order

Placement Real-

Life Example:

Elijah places an order for **1 BMW X5** (Price ₹50,00,000).

In Salesforce:

- A new record is created in the **Vehicle_Order__c** object.
- **Apex Trigger (VehicleOrderTrigger):**
 - Automatically calculates **Total Amount = Price × Quantity**.
 - Checks **stock availability**:
 - If stock > 0 → **Status__c = Confirmed**.
 - If stock = 0 → **Status__c = Pending**.

4. Inventory Update

Real-Life Example:

When Elijah's order is confirmed, the BMW stock decreases by 1 (from 5 → 4).

In Salesforce:

- **Apex Trigger (VehicleOrderTriggerHandler):**
 - Updates **Vehicle__c.Stock_Quantity__c** by subtracting the quantity ordered.
 - Prevents stock from going below zero.

5. Pending → Confirmed Orders (Stock

Refill) Real-Life Example:

If BMW stock is 0 and Elijah's order was pending, when the admin refills **5 BMWs**, the system **automatically confirms pending orders** (if stock is available).

In Salesforce:

- **VehicleTrigger (After Update):**
 - Detects stock refill.
 - Calls `confirmPendingOrders()` in the **VehicleOrderTriggerHandler**.
 - Updates all pending orders to **Confirmed** until stock runs out.

6. Scheduled Batch

Processing Real-Life

Example:

At the end of each day, a **batch job** runs to update **any remaining pending orders** automatically.

In Salesforce:

- **VehicleOrderBatch & VehicleOrderBatchScheduler:**
 - Confirm pending orders if stock is available.
 - Sends admin a log of processed orders.

7. Email Notifications (Flow)

Real-Life Example:

When an order is **confirmed** or **pending**, Elijah gets an **email**

notification. In Salesforce:

- **Record-Triggered Flow:**
 - Sends an email alert:
"Your BMW X5 order is confirmed!"
or
"Your order is pending due to low stock."

8. Users and

Roles Real-Life

Example:

- **Niklaus Mikaelson** (Sales Manager) handles customer orders.
- **Kol Mikaelson** (Inventory Manager) manages vehicle stock.

In Salesforce:

- Roles and Profiles:
 - **Sales Role:** Access to Vehicle_Order__c and Vehicle_Customer__c.
 - **Inventory Role:** Access to Vehicle__c and stock management.

9. Real-Life Benefit

This system ensures:

- **No manual stock checks** (triggers handle it automatically).
- **Automatic order status updates** (Confirmed or Pending).
- **Smooth communication with customers** via email alerts.
- **Dealer assignment** (optional via Flow) ensures orders are assigned to the nearest dealer.

SCREENSHOTS

The screenshot shows the 'Vehicles' page in the WhatNext Vision Motors application. The header includes a search bar and navigation tabs: WhatNext Vision Motors, Vehicle Customers, Vehicle Dealers, Vehicle Orders, Vehicle Service Requests, Vehicle Test Drives, Vehicles (selected), and More. A notification banner at the top states 'Vehicle "Yamaha" was created.' The main content area displays the details for a vehicle named 'Yamaha'. The details are organized into two columns: the left column lists attributes like Vehicle Name, Vehicle Model, Stock Quantity, Price, Vehicle Dealer, Status, and Created By; the right column lists Owner, Last Modified By, and a link to activate Windows. The vehicle details are as follows:

Attribute	Value
Vehicle Name	Yamaha
Vehicle Model	Sedan
Stock Quantity	100
Price	\$80,000
Vehicle Dealer	CP
Status	Available
Created By	raid rafif aufa, 1/6/2026, 4:51 AM
Owner	raid rafif aufa
Last Modified By	raid rafif aufa, 1/6/2026, 4:51 AM

Buttons for 'New Contact', 'Edit', and 'New Opportunity' are visible in the top right corner.

Fig: Vehicles in WhatNext Vision Motors

The screenshot shows the 'Vehicle Customers' page in the WhatNext Vision Motors application. The header includes a search bar and navigation tabs: WhatNext Vision Motors, Vehicle Customers (selected), Vehicle Dealers, Vehicle Orders, Vehicle Service Requests, Vehicle Test Drives, Vehicles, and More. A notification banner at the top states 'Vehicle "Maia" was created.' The main content area displays the details for a vehicle customer named 'Maia'. The details are organized into two columns: the left column lists attributes like Vehicle Customer Name, Email, Phone, Address, Preferred Vehicle Type, and Created By; the right column lists Owner, Last Modified By, and a link to activate Windows. The customer details are as follows:

Attribute	Value
Vehicle Customer Name	Maia
Email	raidraffa@upi.edu
Phone	1245678
Address	ngapak
Preferred Vehicle Type	Sedan
Created By	raid rafif aufa, 1/6/2026, 4:48 AM
Owner	raid rafif aufa
Last Modified By	raid rafif aufa, 1/6/2026, 4:48 AM

Buttons for 'New Contact', 'Edit', and 'New Opportunity' are visible in the top right corner.

Fig: Customers in WhatNext Vision Motors

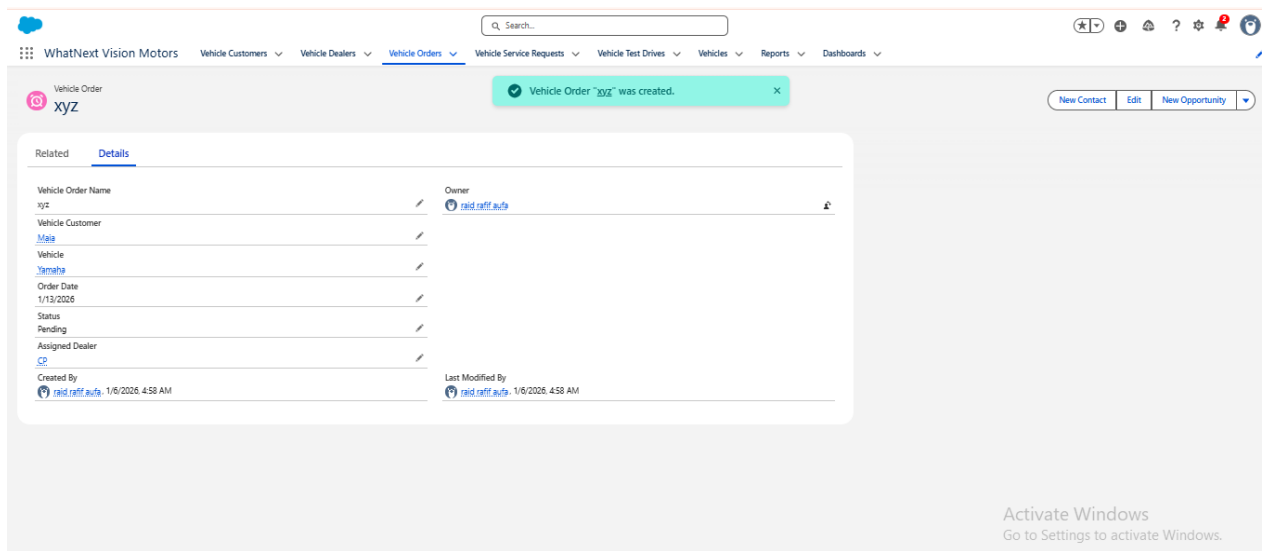


Fig: Vehicle Orders in WhatNext Vision Motors

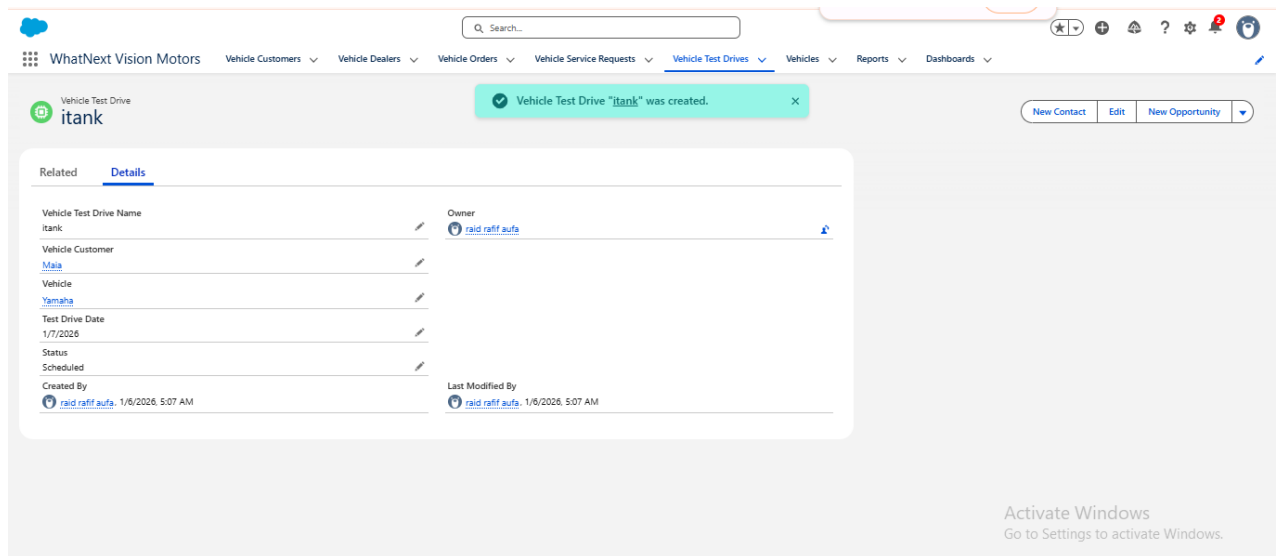


Fig: Test Drives in WhatNext Vision Motors

CONCLUSION

Salesforce is a web-based system for managing customer relationships that helps companies handle their interactions, information, and activities effectively. In this project, it serves the purpose of storing and organizing information about vehicles, dealership sites, and customer requests. The system facilitates automation via Flows, Apex triggers, and batch jobs to guarantee precise inventory checks and intelligent dealer allocations. By utilizing Salesforce, WhatsNext Vision Motors improves its operational productivity and provides a smooth, customer-focused ordering process.

Future Scope

1. Customer Portal Integration

- Develop a **Salesforce Community Portal** where customers can log in to:
 - View their order history and current order status.
 - Check loyalty program points and rewards.
 - Raise queries or service requests directly.

2. Mobile App using Salesforce Mobile SDK

- Build a **mobile application** for store staff and managers to:
 - Manage inventory (add or update stock).
 - Process customer orders in real-time.
 - Access dashboards and order reports on the go.

3. Reports & Dashboard

- Create **dynamic dashboards** and reports in Salesforce for:
 - Real-time vehicle stock levels.
 - Sales trends and revenue analysis.
 - Monitoring loyalty program metrics.

4. AI-Powered Recommendations (Einstein AI)

- Integrate **Salesforce Einstein** to:
 - Analyze customer purchase history.
 - Provide **personalized vehicle recommendations** (e.g., suggest SUVs for customers buying family cars).

5. WhatsApp/SMS Integration

- Use Salesforce **Digital Engagement** or third-party tools (like Twilio) to:
 - Send **instant notifications** about order confirmation or pending status.
 - Update customers about loyalty status or promotional offers.